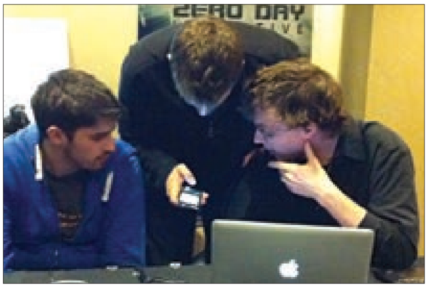
are required to sign a non-disclosure agreement, with the bugs being reported exclusively to the vendors.

In previous Pwn2Own competitions, the latest version of the browser and operating system were used. But this time around the latest browser and OS versions were selected and 'frozen'. Exploits had to be carried out on those. Apple seemingly unaware of the new rules, still went ahead and released Safari 5.0.4 one day before the contest, an update that patched more than 60 security holes. However, the new rules also say that in order to win the prize money of a successful exploit, it would have to also work on the latest version available at the time of the competition.

On Day One of Pwn2Own, there were three attempts lined up, white-hat hackers trying to take out Safari, IE8, and Chrome. The first to fall was Safari 5.0.3m running on a patched Mac OS X 10.6.6. VUPEN, a



Vincenzo Iozzo (left), Pwn2Own official Aaron Portnoy and Willem Pinckaers exploiting the BlackBerry

French security firm was behind the attack, taking over the browser and its hardware five seconds after it had landed on its specially engineered web page. It proved it had attacked the hardware by writing a file to hard disk – effectively bypassing the sandbox. The attack also worked on the latest 5.0.4 version.

After the fact, VUPEN co-founder Chaouki Bekrar said his team's attack was not the simplest to engineer, simply because the number of documented exploit techniques for 64-bit Safari are very low, as it is a platform that requires unusual implementations of normal security bypass techniques. The three-man team apparently took two weeks to assemble the attack.

Next up, Harmony Security assaulted 32-bit Internet Explorer 8 running on 64-bit Windows 7 SP1, taking advantage of three separate vulnerabilities of the browser. Two achieved code execution inside the browser, and the third bypassed the 'Protected Mode' sandbox of IE8. The attack was a one-man effort, with security researcher Stephen Frewer taking nearly five weeks to prepare it.

The attack on Chrome wasn't carried out on Day One, with the scheduled contestants not showing up. Chrome of course is known for its lack of vulnerabilities, and Google went ahead and fixed 24 more flaws a day ahead of the competition – effectively removing all hopes of the participant winning prize money by hacking the latest version as well.

Day Two had tests scheduled for Mozilla Firefox, and native browsers of all the major mobile operating systems - Android, Blackberry OS, iOS, Symbian, and Windows Phone 7. Just like Chrome, it seemed Mozilla had successfully patched Firefox versions 3.5.17 and 3.6.14 before the competition, as the hacker once again was a no-show.

The browsers of the iPhone 4 and then the BlackBerry Torch 9800, running the latest OS versions (iOS 4.2.1 and BlackBerry OS 6.0.0.246), were successfully exploited later in the day, with the hackers required to gain the same level of control over the software and hardware of the phone.

The Nexus S was the scheduled Android target in the competition, and the Dell Venue Pro was picked for Windows Phone 7. Both scheduled attempts were no shows, again.

To note however, are words from last year's Internet Explorer hacker, Peter Vreugdenhil: "The survival of a target at Pwn2Own does not automatically declare it safer than a target that went down." According to Vreugdenhil, there are numerous factors determining the difficulty of a particular hack, everything from the safety of the software itself, to the list of known vulnerabilities and the amount of research that has already been conducted. As an example of this logic, Vreugdenhil added: " Chrome has the advantages of having multiple exploit-mitigation techniques that certainly make it more difficult to hack. As for Android, we see no particular reason why Android would be harder to hack than one of the other targets." d